Smooth Distribution and Depth-Focused Distillation-Based Class-Incremental Learning for SAR Target Detection

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***Abstract*— Incremental learning methods can overcome the problem of catastrophic forgetting in synthetic aperture radar (SAR) target detection models when continuously learning new class data. However, due to the characteristics of SAR images, such as large background variations and relatively stable scale differences between different target classes, existing advanced incremental detection methods perform poorly in the SAR domain. Specifically, current incremental detection research lacks adaptability to spatial distribution changes and insufficiently focuses on the model's localization knowledge, making it significantly limited when addressing the aforementioned SAR characteristics. To tackle these issues, we propose a class-incremental learning method for SAR target detection based on smooth distribution and depth-focused knowledge distillation (SDDFD). First, we design a spatial dimension-based smooth distribution distillation (SDDL) method, which evaluates the effectiveness of spatial response points through concentration assessment and dynamically assigns weights to adapt to different background spatial distributions. Then, based on SDDL, we propose an embeddable depth-focused distillation (DFDL) approach. This approach innovatively enhances shallow localization knowledge and deep classification knowledge from a depth perspective, significantly improving the model's localization ability. Experimental results show that our method outperforms advanced incremental detection methods in various incremental task scenarios, achieving optimal performance. For example, in one-step incremental tasks, the $AP_{0.5}$ of SDDFD compared to the state-of-the-art method ERD, improved by 8.8% and 4.0% on the MSAR-1.0 and SAR-AIRcraft-1.0 datasets, respectively.**

***Index Terms*— Catastrophic forgetting, class-incremental learning, knowledge distillation, synthetic aperture radar (SAR) target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) target detection is a significant research direction in the field of remote sensing, which is widely applied in time-sensitive tasks such as disaster monitoring and marine surveillance. In recent years, deep learning methods based on convolutional neural networks (CNNs) have achieved great success in both SAR target recognition [1], [2], [3] and SAR target detection [4], [5], [6], [7], [8] due to their high detection efficiency and accuracy. In the traditional CNN training paradigm, SAR target recognition or detection models typically require joint training with data from all categories to ensure that the model performs well. However, with the advancement of radar data collection, the acquisition frequency of new class data has increased, posing a new challenging task for SAR target detection research: how to update parameters on models trained with old class data to quickly learn the continuously acquired new class data. Traditional training strategies update the model by retraining with both old and new class data, of which the storage and training time cost will grow linearly as the number of updates increases. Additionally, fine-tuning the old model using only new class data will significantly degrade the detection capability for old classes, a phenomenon known as "catastrophic forgetting" [9], [10]. In this context, incremental learning methods have been proposed to address the challenge of enabling models to incrementally learn new class data while resisting forgetting old classes.

The primary challenge in incremental detection methods lies in retaining the classification and localization knowledge of old class targets. As an early exploration, incremental learning methods were first introduced in the field of target classification [11], [12], [13] and have since achieved success in SAR target incremental recognition [14], [15], where the focus is solely on preserving classification knowledge.

Following the research of LWF, incremental detection methods based on knowledge distillation have been proposed and have become the mainstream approach in this field. The ILOD [16] innovatively applied the LwF to the Fast RCNN [17] detector, enabling the model to resist forgetting. Following this, various knowledge distillation-based incremental detection methods were proposed [18], [19], [20] and achieved success in resisting forgetting. However, most existing incremental detection methods assume that the detection

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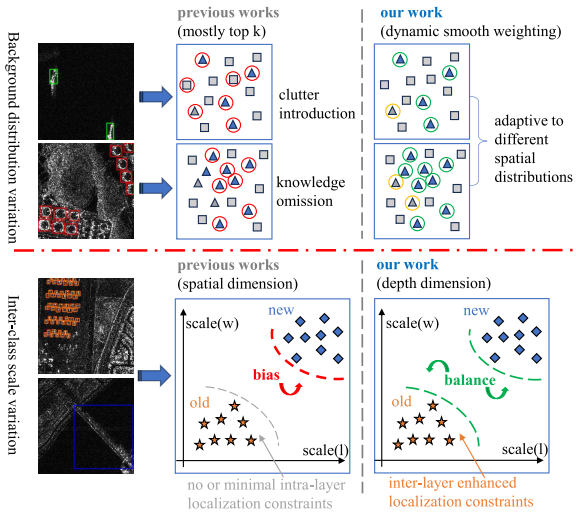


Fig. 1. Examples of challenges in incremental detection for SAR targets. Targets in each scene are marked with rectangular boxes, with different colors corresponding to different classes. Specifically, green, red, orange, and blue boxes indicate ships, oil tanks, airplanes, and bridges, respectively.

model's localization ability is independent of the target class, thus paying little attention to the preservation of localization knowledge. Additionally, the strategies of these methods for selecting effective knowledge in different class backgrounds are mostly based on fixed thresholds, leading to poor adaptability when interclass background differences are significant.

The above issues result in the limited performance of existing advanced incremental detection methods when addressing the following two characteristics of SAR images.

First, due to the large field of view in SAR imaging, the scenes of different target classes correspond to different terrains, resulting in significant background variations. For instance, the background for oil tanks in Fig. 1 is bright land, while the background for ships is the dark sea. This leads to differences in the quantity and spatial distribution of effective knowledge under different backgrounds during knowledge distillation. Previous works [18], [21], [22] relied on fixed thresholds or fixed quantities to extract effective knowledge from the spatial dimension. However, these approaches can result in the omission of knowledge or the introduction of excessive noise when dealing with varying SAR background distributions, thus presenting significant limitations.

Second, due to the long imaging distance of SAR, the relative distances between targets of different categories and the imaging platform have minimal impact on the imaging scale. As a result, the scale information for each class of target is concentrated in a specific range, and the scale relationship between different categories strictly adheres to their physical size relationships, meaning that localization knowledge becomes associated with the target class. For instance, Fig. 1 shows that airplanes are generally small-scale, ships and oil tanks are medium-scale, and bridges are always large-scale. When there is a significant scale difference between new and old target classes during the incremental learning process, the model's localization ability will be biased to the new scale. Existing methods [18] and [22] assumed similar scale information for each target class and did not specifically review localization knowledge. Methods [19] and [20] considered localization knowledge but simply replicated the approach used for preserving classification knowledge, neglecting the impact differences of these two types of knowledge across different depth layers. This results in inefficient retention of localization knowledge. In general, these methods assume that each target class has rich multiscale information and generally lacks attention to localization knowledge, resulting in significant localization capability loss in SAR scenarios when there exits large interclass-scale differences.

Inspired by the knowledge distillation approach [12], [19] and addressing the characteristics of SAR images, this article proposes a smooth distribution and depth-focused distillation-based (SDDFD) class-incremental learning method for SAR target detection. Specifically, our method is divided into two modules: smooth distribution distillation (SDDL) and embeddable depth-focused distillation (DFDL). The SDDL addresses the issue of the large interclass background differences from the spatial dimension. The DFDL module is inspired by the FPN concept, which highlights that different depths of feature layers have varying importance for localization and classification [23]. Based on this, it improves localization capabilities from the depth dimension to tackle the issue of correlation between localization knowledge and target classes. The main contributions of this article are summarized as follows.

1) We design a new function named response concentration assessment (RCA) for evaluating the effectiveness of each response point in the model's head layers.
2) To address the issue of SAR background variations, an SDDL module based on the spatial dimension is proposed. This module dynamically applies soft weight enhancement to response points based on their concentration, adapting to different background distributions.
3) To address the issue of the correlation between localization knowledge and SAR target classes, an embeddable DFDL module is proposed. This module refines effective localization and classification regions by considering the depth relationships of different feature layers and response layers, allocating importance accordingly, thereby enhancing the high-precision localization capability of the incremental model.

II. Related Work

A. Incremental Learning

Incremental learning methods have been proposed to address the issue of catastrophic forgetting when models incrementally learn new classes. Among these, class-incremental learning [24], [25] is currently a hot research topic.

Class-incremental learning methods were initially explored in the domain of target recognition, where the primary focus is on retaining classification knowledge of old classes. The main paradigms for incremental classification research include data replay [11], [26], [27], generative replay [28], [29], and regularization [12], [30], [31]. Following these paradigms, data replay-based [15], [32], [33], [34] and regularization-based [35], [36], [37] class-incremental learning methods have

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rapidly developed in the field of SAR target recognition. For example, Dang et al. [32] selected samples on the boundaries of the feature space to represent the distribution of old class data, helping the model resist forgetting. Gao et al. [15] used a herding algorithm to select representative old-class samples for incremental training and designed an intraclass clustering loss to preserve the knowledge of the old classes. Yu et al. [37] designed a multilevel adaptive knowledge distillation network to retain the model's recognition ability for old classes.

Overall, incremental classification methods are relatively flexible and diverse since they only need to focus on retaining classification knowledge, and regularization-based methods have been receiving increasing attention.

B. Incremental Detection

Compared with incremental classification, incremental detection methods face the additional challenge of preserving localization knowledge, making the research more complex. Consequently, studies on incremental detection methods have started later and are especially less developed in the field of SAR target detection.

Currently, knowledge distillation-based regularization methods are the primary research direction. Existing incremental detection studies address the selection of effective knowledge regions from the spatial dimension, with increasing attention paid to localization knowledge. Since localization knowledge in object detection models is difficult to evaluate, early studies generally avoided focusing on the evaluation of localization knowledge. For instance, the ILOD [16] and Faster-ILOD [22] methods use class-agnostic proposals to replace localization knowledge, exploring incremental learning methods for two-stage networks. The SID method [18] argues that localization responses have little impact on performance, thus only using all spatial points from intermediate feature layers and classification response layers for distillation. The RILOD method [21] selects all spatial points from intermediate feature layers and classification response layers and chooses the top-k localization response points with the highest classification scores for distillation. Later, the introduction of the GFL localization representation method [38] provided technical support for evaluating localization knowledge. Benefiting from this, the ERD method [19] separately evaluates and selects effective classification and localization knowledge for distillation. Tian et al. [20] conducted pioneering research in SAR incremental detection, which selected the top N proposal regions from the classification and localization response layers based on feature layer scores for distillation.

Overall, current incremental detection methods rely heavily on fixed thresholds set empirically for selecting effective regions in the spatial dimension, which may limit performance in SAR scenarios with different background distributions. Furthermore, existing incremental detection methods pay little attention to localization knowledge, even though preserving localization knowledge has always been a challenging aspect that distinguishes incremental detection from incremental classification. Works that include localization knowledge [19], [20], [21] focus solely on the quality relationships between points within each layer (referred to as spatial dimension relationships in this article) and lack consideration of interlayer relationships (referred to as depth dimension relationships in this article). These approaches for preserving localization knowledge simply mimic that for classification knowledge, leading to poor specificity and efficiency, and leaving significant room for improvement in localization capabilities.

III. Proposed Method

In this section, we will introduce the proposed method in five subsections: the overall structure, the SDDL module, the hierarchical response focal (HRF) module, the depth feature decoupling (DFD) module, and the overall loss. The HRF module and the DFD module together form the DFDL method.

A. Overall Structure

The overall framework of the SAR target class-incremental detection method proposed in this article is illustrated in Fig. 2. The basic structure of this method follows the knowledge distillation paradigm, where a teacher–student model is constructed. The teacher model is the old class model with frozen parameters, while the student model is initialized with the parameters of the old class model but remains unfrozen. Both the teacher and student models use GFL [38] with FPN [23] as the detector. The teacher model guides the training of the student model by generating distillation loss, and the trained student model becomes the final incremental detection model.

Specifically, our method consists of three modules. The SDDL and HRF modules are used to generate response distillation loss, while the DFD module is used to generate feature distillation loss. Together, these form the total distillation loss. Since both the HRF and the DFD methods consider the effectiveness of knowledge from the depth dimension, we collectively refer to them as the DFDL method.

To facilitate the description of our method, the concepts of spatial and depth dimensions are defined as follows in this article. In a deep learning network, a feature or response layer can be represented as $F^d \in \mathbb{R}^{h\times w\times c}$ or $R^d \in \mathbb{R}^{h\times w\times c}$. Here, d denotes the depth dimension, representing the relationship between layers in terms of depth—the more downsampling convolutional modules a layer has passed through, the deeper its depth. $h \times w$ represents the spatial dimension of the layer, and c represents the channel dimension.

B. Smooth Distribution Distillation

The SDDL method is a knowledge distillation approach based on dynamically assigning weights to the effectiveness of response points in the head layer according to the spatial dimension. This method addresses the issue of significant background variations among different SAR target classes and the varied spatial distribution of effective knowledge, avoiding the drawbacks of previous methods that rely on fixed thresholds set by experience. The overall structure is shown in Fig. 3. This method first uses the RCA function to evaluate the effectiveness of classification and localization response points. Then, it uses a soft response enhancement (SRE) function to hierarchically weight the effective response points of each

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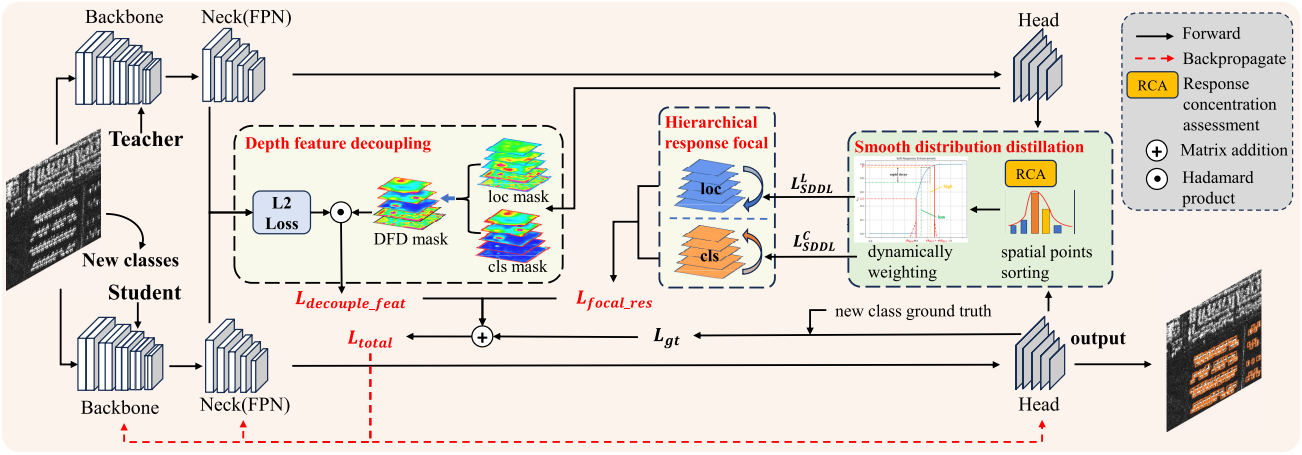


Fig. 2. Overall structure of SDDFD. Our method comprises three modules: SDDL, DFD, and HRF. The combination of SDDL and HRF generates response constraints, while DFD generates feature constraints. These constraints are jointly involved in the backpropagation of the student model's parameters, guiding it to retain old class knowledge during updates. Upon completion of the training with only new class data, the student model will possess the capability to detect both new and old classes and output predictions accordingly.

layer, achieving the goal of adaptively extracting effective knowledge while suppressing noise.

1) Response Concentration Assessment: In a one-stage detection network, the head layer consists of classification and localization responses. The GFL [38] innovatively discretizes the localization response into probabilities, allowing it to share a similar evaluation and distillation approach as the classification response [39]. Therefore, we designed an RCA function to evaluate the effectiveness of both types of responses, which is

$$f_{\mathrm{RCA}}\left(R_j^{d_i}\right)=\sum_{k=1}^{n}\left(R_{j,k}^{d_i}\right)^2 \tag{1}$$

where $R_j^{d_i}\in\mathbb{R}^n(j\in 1,\ldots,\ h_i\times w_i|n=c\cdot\mathrm{or}\cdot b)$ represents the probability prediction vector of the jth response point in the d_ith layer, and $R_{j,k}^{d_i}$ denotes the kth predicted probability value of $R_j^{d_i}$, and $\sum_{k=1}^{n}(R_{j,k}^{d_i})=1$. In classification response points, $R_{j,k}^{d_i}$ corresponds to the predicted probability of the kth class target, where $n=c$. In localization response points, $R_{j,k}^{d_i}$ corresponds to the probability of the bounding box length being k, where $n=b$. The RCA function reflects the degree of numerical dominance of the top categories (i.e., one or more categories with the highest predicted values) in the prediction vector compared to other smaller predicted categories. The higher the RCA value, the more concentrated the model's predicted knowledge is in the top categories, indicating higher knowledge distinguishability and greater effectiveness for that response point.

Additionally, the RCA method, like the traditional maximum value-based evaluation method, considers a response point with a maximum prediction value of 1 to be the most effective, and a response point with a maximum value of $1/n$ to be the least effective. However, apart from these two extreme points, when multiple response points have equal maximum values in their prediction vectors, the traditional maximum value-based evaluation method cannot differentiate their effectiveness. In contrast, the RCA method can distinguish them by considering the distribution of nonmaximum values in the prediction vector.

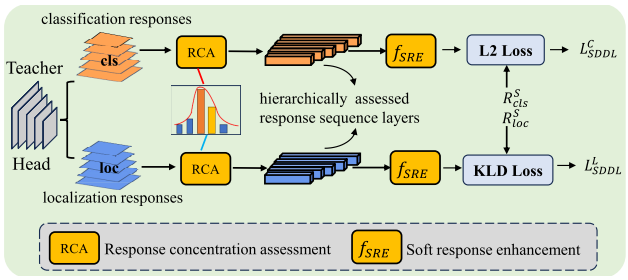


Fig. 3. Overall structure of the smooth distribution distillation method.

2) Soft Response Enhancement: Due to the FPN [23] structure adopted in the neck layer of the detector, both the classification response and the head response each have 5 layers, denoted as $d_0\sim d_4$. The SRE function dynamically assigns weights to the response points of the five layers based on the concentration of each layer's response points. Specifically, inspired by ATSS [40], we use the statistical distribution characteristics of each layer's responses R^{d_i} to generate the concentration threshold $\mathrm{th}_{\mathrm{RCA}}^{d_i}$ for the d_i layer. The threshold is calculated as

$$\mathrm{th}_{\mathrm{RCA}}^{d_i}=\mathrm{mean}\left(f_{\mathrm{RCA}}\left(R^{d_i}\right)\right)+\mathrm{std}\left(f_{\mathrm{RCA}}\left(R^{d_i}\right)\right). \tag{2}$$

Since this threshold is only related to the statistical distribution of the current scene, it can adapt to changes in the background distribution of different SAR target classes. Response points with concentration values below the threshold are considered outliers and are assigned a weight of 0. For any response point $R_j^{d_i}$ with concentration value above the threshold, we use the SRE function to dynamically assign weight to it, which is

$$f_{\mathrm{SRE}}\left(R_j^{d_i}\right)=\frac{1}{1+e^{-\gamma\left(\frac{f_{\mathrm{RCA}}\left(R_j^{d_i}\right)}{\mathrm{th}_{\mathrm{RCA}}^{d_i}}-1\right)}} \tag{3}$$

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where the variable γ is used to control the softness of the response enhancement from $\mathrm{th}_{\mathrm{RCA}}^{d_i}$ to a high-quality concentration $\mathrm{RCA}_{\mathrm{high}}^{d_i}$, where

$$\mathrm{RCA}_{\mathrm{high}}^{d_i} = \mathrm{th}_{\mathrm{RCA}}^{d_i} + \mathrm{std}\left(f_{\mathrm{RCA}}\left(R^{d_i}\right)\right). \tag{4}$$

Setting the soft enhance value of $\mathrm{RCA}_{\mathrm{high}}^{d_i}$ point to hyperparameter p ($p = 0.9$ by default), the variable γ can be regulated as

$$\gamma = \frac{(\ln(p) - \ln(1-p)) \times \mathrm{th}_{\mathrm{RCA}}^{d_i}}{\mathrm{std}\left(f_{\mathrm{RCA}}\left(R^{d_i}\right)\right)}. \tag{5}$$

The concave mathematical characteristic of the SRE function ensures that, during the weighting process, the weight of response points near $\mathrm{RCA}_{\mathrm{high}}^{d_i}$ will decrease slightly, while the weight of response points near $\mathrm{th}_{\mathrm{RCA}}^{d_i}$ will decrease rapidly. This guarantees that the distillation constraint on the responses suppresses noise interference while preserving as much effective old-class knowledge as possible.

The final loss of SDDL for the classification response and the localization response generated by the head of the teacher model are

$$L_{\mathrm{SDDL}}^{L}\left(R_{\mathrm{loc}}^{T}, R_{\mathrm{loc}}^{S}\right) = \sum_{d_i=d_0}^{d_4} \sum_{j=1}^{h_i \times w_i} f_{\mathrm{SRE}}\left(R_{\mathrm{loc},j}^{T,d_i}\right)\left(\sum_{e \in E} L_{\mathrm{KL}}\left(R_{\mathrm{loc},j}^{T,d_i}, R_{\mathrm{loc},j}^{S,d_i}\right)\right) \tag{6}$$

and

$$L_{\mathrm{SDDL}}^{C}\left(R_{\mathrm{cls}}^{T}, R_{\mathrm{cls}}^{S}\right) = \sum_{d_i=d_0}^{d_4} \sum_{j=1}^{h_i \times w_i} f_{\mathrm{SRE}}\left(R_{\mathrm{cls},j}^{T,d_i}\right)\left(R_{\mathrm{cls},j}^{T,d_i,j} - R_{\mathrm{cls},j}^{S,d_i}\right)^2 \tag{7}$$

where T and S represent the teacher and student model, respectively. The loc and cls represent the localization and classification response, respectively. E stands for the four edges of the predicted localization bounding box. L_{KL} stands for Kullback–Leibler divergence loss.

C. Hierarchical Response Focal Distillation

Previous incremental detection studies have only selected effective knowledge from the spatial dimensions of a given layer, assuming that the feature maps of each layer are equally important. However, research on the FPN [23] has shown that there are different resolutions and significant semantic gaps between feature layers of varying depths. Specifically, features at higher semantic levels are helpful for classification knowledge, while features with higher resolution are helpful for localization knowledge. Therefore, previous studies have overlooked the impact of the depth differences between layers, which limits the distinction and utilization of effective knowledge. Inspired by FPN research, we designed the HRF method, aiming to assign importance levels to the localization and classification responses of different layers from a depth dimension, further enhancing the model's retaining of localization capability and thereby addressing the challenge of learning scale differences among different SAR target classes.

Specifically, the depth of feature maps is determined by the number of downsampling operations they have undergone. The more downsampling operations, the deeper the feature map's depth, which corresponds to a higher semantic level and lower resolution. Therefore, in the five feature layers d_0 to d_4 of the FPN, the depth progressively increases from d_0 to d_4. The five layers of localization and classification response maps, computed from the FPN features, maintain the same depth relationship from d_0 to d_4 as the FPN. The HRF method designs the following hierarchical depth-weighting strategy based on the scale of each layer, diminishing the model's attention to shallow classification responses and deep localization responses, thereby indirectly strengthening deep classification knowledge and shallow localization knowledge. The focus weights for each layer of classification and localization response maps are as follows:

$$fc_{\mathrm{cls}}^{d_i} = \frac{h_4 \times w_4}{h_i \times w_i} \tag{8}$$

and

$$fc_{\mathrm{loc}}^{d_i} = \frac{h_i \times w_i}{h_0 \times w_0} \tag{9}$$

where $h_i \times w_i$ represents the scale of the response map at the d_i layer, respectively. The purpose of using scale control weights for each layer is to prevent the sum of distillation losses from nonessential layers from overwhelming the distillation loss from important layers.

By combining the HRF method with the SDDL, the final deep focal response loss is defined as

$$L_{\mathrm{focal_res}} = \sum_{i=0}^{4}\left(fc_{\mathrm{loc}}^{d_i} \times L_{\mathrm{SDDL}}^{L}\left(R_{\mathrm{loc}}^{T,d_i}, R_{\mathrm{loc}}^{S,d_i}\right)\right) + \sum_{i=0}^{4}\left(fc_{\mathrm{cls}}^{d_i} \times L_{\mathrm{SDDL}}^{C}\left(R_{\mathrm{cls}}^{T,d_i}, R_{\mathrm{cls}}^{S,d_i}\right)\right). \tag{10}$$

The HRF method is a plug-and-play module that is not only applicable to the SDDL method presented in this article but can also be embedded in any incremental detection method that processes based on hierarchical spatial dimensions.

D. Depth Feature Decoupling

The DFD method is distilled at FPN features, which assigns different focus regions to feature layers from the depth dimension. Its primary functions are twofold: first, it enhances the attention of shallow features to localization knowledge areas, improving the model's localization capability. Second, it uses neck feature layers to supplement an appropriate amount of intermediate layer distillation knowledge, avoiding the insufficient knowledge volume caused by long-range regularization when only response layers are distilled, thereby enhancing the robustness of the incremental detection algorithm.

The overall framework of the DFD is shown in Fig. 4. This method consists of two parallel branches: the lower branch performs hierarchical channel attention enhancement, and the upper branch conducts DFD.

The hierarchical channel attention enhancement in the lower branch mainly aims to reduce channel noise interference.

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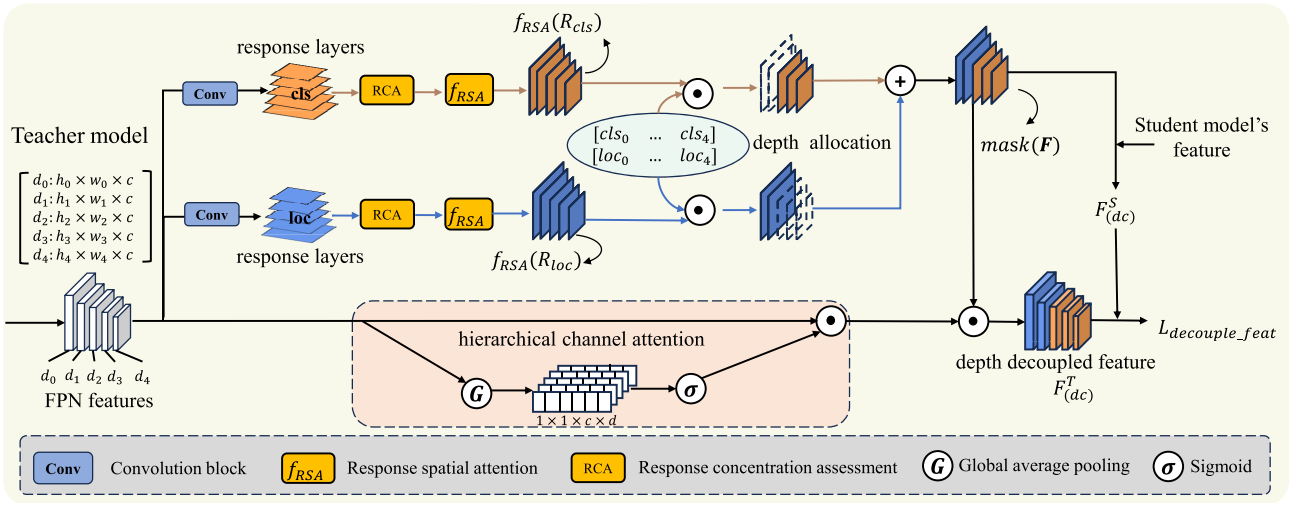


Fig. 4. Framework of DFD. This framework consists of two parallel branches. The lower branch uses a hierarchical channel attention mechanism to reduce channel noise. The upper branch performs DFD by assigning attention masks generated from localization or classification responses to different feature layers of the FPN based on their depth, allowing shallow layers to focus on localization and deep layers to focus on classification. This effectively decouples the two types of knowledge from overlapping features along the depth dimension.

We represent the feature F^{d_i} of the d_i layer in the FPN as a set of vectors, namely $F^{d_i} = \{F_1^{d_i}, \ldots, F_n^{d_i}\}$, $F_k^{d_i} \in \mathbb{R}^c$, and $n = h_i \times w_i$. The global average pooling for this layer can be represented as

$$f_{\text{gp}}\left(F^{d_i}\right) = \frac{\sum_{k=0}^{h_i \times w_i}\left(F_k^{d_i}\right)}{h_i \times w_i}. \tag{11}$$

Then, the feature of d_i layer after channel attention can be

$$F_{(\text{ca})}^{d_i} = F^{d_i} \cdot \text{sigmoid}\left(f_{\text{gp}}\left(F^{d_i}\right)\right). \tag{12}$$

The DFD method generates the knowledge regions that each feature layer in the neck layer needs to focus on in the upper branch based on the response maps of the head layers. Specifically, we first use the RCA function to calculate the concentration of each point in the classification and localization responses and select the response points with higher concentration in each layer as the knowledge regions that need to be focused on. Through this process, spatial attention masks for classification knowledge and localization knowledge are generated separately for all five layers. The spatial attention masks for the classification and localization knowledge of the jth point in the d_ith layer are calculated as follows, respectively:

$$f_{\text{RSA}}\left(R_{\text{cls},j}^{d_i}\right) = \begin{cases} 1, & \text{if } f_{\text{RCA}}\left(R_{\text{cls},j}^{d_i}\right) \geq \text{th}_{\text{RCA,cls}}^{d_i} + \text{std}\left(f_{\text{RCA}}\left(R_{\text{cls}}^{d_i}\right)\right) \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

and

$$f_{\text{RSA}}\left(R_{\text{loc},j}^{d_i}\right) = \begin{cases} 1, & \text{if } f_{\text{RCA}}\left(R_{\text{loc},j}^{d_i}\right) \geq \text{th}_{\text{RCA,loc}}^{d_i} + \text{std}\left(f_{\text{RCA}}\left(R_{\text{loc}}^{d_i}\right)\right) \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

After obtaining the response spatial attention masks, we construct a depth assignment vector to allocate different attention regions to the five FPN feature layers. Specifically, we use the mask for the localization response only in the shallow layers of the feature map, and the mask for the classification response only in the deeper layers. This ensures that shallow layers focus on localization areas, while deep layers focus on classification areas. In this article, the depth allocation vectors are constructed as $[\text{cls}_0,\ \text{cls}_1,\ \text{cls}_2,\ \text{cls}_3,\ \text{cls}_4] = [0,\ 1-ld,\ 1,\ 1,\ 1]$ and $[\text{loc}_0,\ \text{loc}_1,\ \text{loc}_2,\ \text{loc}_3,\ \text{loc}_4] = [1,\ ld,\ 0,\ 0,\ 0]$. As the number of downsampling operations increases when depth goes deeper, the spatial dimensions $h_i \times w_i$ of deeper features become much smaller than those of shallower features. To ensure that the student model acquires an adequate amount of classification knowledge, we allocate the deep features at depths d_2 to d_4 to the review of classification knowledge, and the shallow features at depth d_0 to the review of localization knowledge, thereby balancing the volume of both types of knowledge. The allocation for layer d_1 is flexibly determined by the relative standard deviation of the concentration values in the d_0 and d_1 layers. The value of ld to control the allocation for layer d_1 can be calculated as

$$ld = \begin{cases} 1, & \text{std}\left(f_{\text{RCA}}\left(R_{\text{loc}}^{d_1}\right)\right) > \text{std}\left(f_{\text{RCA}}\left(R_{\text{loc}}^{d_0}\right)\right) \\ 0, & \text{std}\left(f_{\text{RCA}}\left(R_{\text{loc}}^{d_1}\right)\right) \leq \text{std}\left(f_{\text{RCA}}\left(R_{\text{loc}}^{d_0}\right)\right). \end{cases} \tag{15}$$

After depth allocation, the concatenated hierarchical spatial response attention mask for feature layer d_i is as follows:

$$\text{mask}\left(F^{d_i}\right) = \text{cls}_i \cdot f_{\text{RSA}}\left(R_{\text{cls},j}^{d_i}\right) + \text{loc}_i \cdot f_{\text{RSA}}\left(R_{\text{loc},j}^{d_i}\right). \tag{16}$$

Finally, the depth-decoupled feature for the d_i layer, after combining the upper and lower branches, is as follows:

$$F_{(\text{dc})}^{d_i} = \text{mask}\left(F^{d_i}\right) \cdot F_{(\text{ca})}^{d_i}. \tag{17}$$

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Then, the feature distillation loss for DFD can be

$$L_{\text{decouple_feat}} = \sum_{i=0}^{4} \sum_{j=1}^{h_i \times w_i} \left(F^{T,d_i}_{(\text{dc}),j} - F^{S,d_i}_{(\text{dc}),j} \right)^2. \tag{18}$$

Due to the involvement of feature layers in both localization and classification response generation in one-stage detection networks, each layer inherently contains mixed regions of classification and localization knowledge. The DFD method addresses this by allocating specific knowledge focus regions to different depth feature layers. This decouples localization and classification knowledge from the depth dimension, refining the incremental model's ability to distinguish and learn from both types of knowledge, and thereby avoiding the introduction of excessive noise due to knowledge overlap.

E. Overall Loss

To sum up, the total loss used to train the student model is as follows:

$$L_{\text{total}} = L_{\text{gt}} + \lambda \cdot \left(L_{\text{focal_res}} + L_{\text{decouple_feat}} \right) \tag{19}$$

where L_{gt} represents the loss used to guide the student model in learning new class targets, utilizing the ground truth from new data. $L_{\text{focal_res}}$ and $L_{\text{decouple_feat}}$ together constitute the distillation loss, which is used to review old class knowledge. λ is a hyperparameter to balance the constraints between the new and old classes ($\lambda = 0.25$ by default).

IV. Experiment

In this section, we conduct experiments using the MSAR-1.0 [41] and SAR-AIRcraft-1.0 datasets [42] to validate the performance of our method. First, we introduce the details of the datasets and performance evaluation metrics. Then, we explain the settings related to the incremental experiments. Next, we examine the effectiveness of each submethod of SDDFD through ablation experiments. Finally, we compare the performance of our method with existing state-of-the-art incremental detection methods for one-stage networks, demonstrating the superiority of our approach.

A. Datasets and Evaluation Metrics

1) Datasets: To validate the performance of our method, we conducted experiments using the MSAR-1.0 [41] and SAR-AIRcraft-1.0 [42] datasets. The MSAR-1.0 dataset, released in 2022, is a large-scale multiclass SAR target detection dataset comprising four classes of targets. The SAR AIRcraft-1.0 dataset, released in 2023, is a high-resolution SAR aircraft detection and recognition dataset, which includes seven aircraft categories. More information about the two datasets is presented in Table I. The characteristics of these two datasets meet the requirements of our experiments.

In the MSAR-1.0 dataset, there is a significant sample imbalance problem, with 26 094 images for the ship class and only 108 images for the airplane class [41]. To prevent the sample imbalance from affecting the experimental results beyond the scope of this study, we performed data balancing on the MSAR-1.0 dataset. The balanced dataset is shown in Table II. Specifically, we limited and randomly selected a balanced number of targets for each class. It should be noted that our balancing strategy is intended solely to ensure the detection model achieves good performance, rather than representing the optimal data balancing solution.

TABLE I
Information About the Datasets

MSAR-1.0	
Resolution	5m,3m
Image Size (pixel)	256×256, 2048×2048
Categories	ship, oil tank, airplane, bridge
Images	28449
Targets	60396
SAR-AIRcraft-1.0	
Resolution	1m
Image Size (pixel)	800×800, 1000×1000, 1200×1200, 1500×1500
Categories	A220, A320/321, A330, ARJ21, Boeing 737, Boeing 787, others
Images	4368
Targets	16463

TABLE II
Balanced Quantities of the MSAR-1.0 Dataset

	ship	oil tank	airplane	bridge
Images(original)	26094	1248	108	1582
Targets(original)	39858	12319	6368	1851
Images(balanced)	2023	553	101	1253
Targets(balanced)	3001	5005	5699	1390

2) Evaluation Metrics: We use $AP_{0.5}$ as the primary performance evaluation metric for this experiment, which is also one of the main evaluation standards commonly used in existing target incremental detection methods.

Average precision (AP) is the area under the precision–recall (P–R) curve bounded by the precision and recall axes. The higher the AP value, the closer the area under the P–R curve is to 1, indicating better detection performance of the model.

AP_α represents the AP value at the intersection-over-union (IoU) threshold of α. The higher the α value, the greater the challenge to the model's localization capability. In this experiment, we also record the performance of each method at $AP_{0.75}$ and $AP_{0.5:0.95}$ to verify the effectiveness of our method under high localization requirements.

B. Experimental Setting

1) Implementation Detail: In this article, we use the GFL-based [38] FCOS [43] network as the detector mode, with ResNet50 as the backbone and FPN [23] as the neck. We initially train the model according to the method described in the original GFL paper to obtain a teacher model trained on old classes. The entire experiment is conducted on PyTorch 1.13, CUDA 11.3, and cuDNN 8.3, utilizing an NVIDIA RTX 4070ti GPU for hardware support.

In the MSAR-1.0 dataset, we split the training and validation sets in a 6:2 ratio. For the SAR-AIRcraft-1.0 dataset,

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TABLE III
COMPARISON OF DETECTION RESULTS UNDER DIFFERENT STRATEGIES OF SMOOTH DISTRIBUTION DISTILLATION

	MSAR-1.0	SAR-AIRcraft-1.0
Method	$AP_{0.5}(2+2)$	$AP_{0.5}(4+3)$
Joint Training	83.7	73.8
Fine-Tune	34.7	34.0
SDDL: Non-hierarchical	66.7	65.9
SDDL: Maximun	69.3	66.9
SDDL: Ours	**71.3**	**67.8**

we follow the official dataset division method, allocating the training and validation sets in a 7:1 ratio. During the training process, we set the hyperparameters p and λ at 0.9 and 0.25, respectively, the batch size is set to 8, the initial learning rate is set at 0.001, and the model is trained for a total of 24 epochs.

2) Incremental Training Strategy: In our class-incremental detection tasks, we employed three different training strategies to update the old model.

a) Joint training: This strategy involves retraining the original model using all available old and new class data. Joint training, while not considered an incremental detection method due to its reliance on full access to old class data, is often viewed as the performance upper bound for all incremental methods. The smaller the performance gap between a given incremental learning method and joint training, the more effective the incremental method is.

b) Fine-tune: This strategy involves freezing the parameters of the head layers related to the old class targets and fine-tuning the old model using only new class data. This strategy generally leads to catastrophic forgetting.

c) Incremental learning: This strategy employs a knowledge distillation-based approach to create constraints for the old tasks, combined with new class data to update the old model.

3) Incremental Task Division: In this experiment, based on the requirements of the incremental task, we divided it into the following two forms.

a) One-step: This approach divides the task into two phases. In the first phase, a subset of categories from the dataset is used as old class data to train an initial model, referred to as the old model. In the second phase, the remaining categories are used as new class data, and different incremental training strategies are applied to update the old model to obtain the final incremental detection model. One-step incremental detection tasks are typically denoted in the form of $a1 + a2$, where $a1$ denotes the number of old class categories in the first phase and $a2$ represents the number of new class categories in the second phase.

b) Multistep: This approach divides the task into multiple phases, each of which follows a similar training philosophy to the one-step incremental task. In the first phase, a subset of categories from the dataset is used as old class data to train an initial model, which serves as the old model. In all subsequent phases, a selection of categories that have not yet been included in the training are chosen as new class data. The incremental detection model completed in the previous phase is treated as the old model and is updated using various incremental training strategies, resulting in a new incremental detection model of the current phase. Multistep incremental detection tasks are usually represented in the form of $a1 + a2 + a3 + \cdots + an$, where ai indicates the number of categories involved in the training data of the ith phase. In this section, we use the format $a1 + a2 + a3$ to complete the multistep incremental task experiments. All experiments in this section were run four times with the same settings, and the final results were averaged.

TABLE IV
RESULTS OF DEPTH-FOCUSED SUBMETHOD ON THE MSAR-1.0 DATASET WITH A 2 + 2 TASK

Method	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.5:0.95}$
Joint Training	83.7	62.3	55.8
SDDL	71.3	49.6	45.2
SDDL+SA	69.6	48.4	43.9
SDDL+DFD	73.8	51.6	46.5
SDDL+HRF	76.9	53.5	48.1
Ours:SDDL+DFD+HRF	**77.5**	**55.9**	**49.2**

TABLE V
RESULTS OF THE DEPTH-FOCUSED SUBMETHOD ON THE SAR-AIRCRAFT-1.0 DATASET WITH A 4 + 3 TASK

Method	$AP_{0.5}$	$AP_{0.75}$	$AP_{0.5:0.95}$
Joint Training	73.8	46.9	45.3
SDDL	67.8	42.0	40.8
SDDL+SA	65.2	37.6	36.6
SDDL+DFD	68.5	43.5	41.2
SDDL+HRF	69.8	44.0	42.3
Ours:SDDL+DFD+HRF	**70.9**	**44.5**	**42.8**

C. Ablation Study

In this section, we consistently employ the one-step incremental task approach to validate the effectiveness of each submethod proposed in our article. We first assess the effectiveness of the smooth distribution distillation and then evaluate the efficacy of the depth-focused distillation method. Finally, we examine the impact of hyperparameters on the performance of our overall method.

1) Effectiveness of SDDL: In this section, we primarily test the performance of the model when only using the SDDL method, to validate the effectiveness of the RCA and the hierarchical response enhancement strategy. As shown in Table III, we use "SDDL" to represent our proposed smooth distribution distillation method. "Maximum" refers to the use of the traditional maximum value assessment method for selecting effective response points in SDDL, instead of RCA. "Nonhierarchical" means the SDDL method without hierarchical assessment, which assesses and enhances the effective response points across all five layers collectively, rather than evaluating each layer independently within its own criteria of the threshold of th_{RCA}. "Ours" indicates the full

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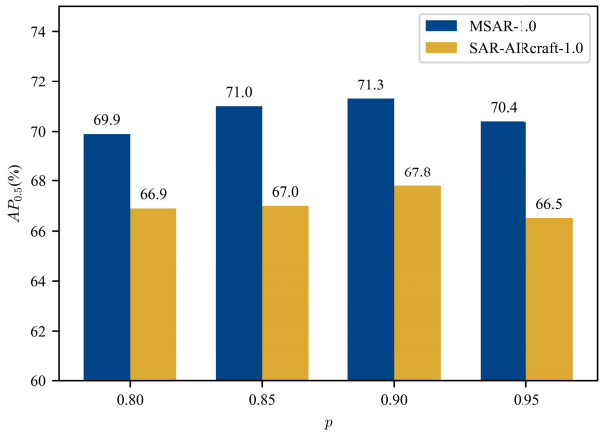


Fig. 5. Experiment results under different values of hyperparameter p. When the value of p is set to 0.9, the performance is optimal.

use of the strategies proposed in SDDL, including RCA and hierarchically SRE.

The results from Table III show that the "Nonhierarchical" strategy yields the poorest training outcomes, achieving $AP_{0.5}$ of only 66.7% and 65.9% on the $2+2$ and $4+3$ tasks of the MSAR-1.0 and SAR-AIRcraft-1.0 datasets, respectively. In comparison, the performance of methods "Maximum" and "Ours," which utilize the hierarchical enhancement strategy, is significantly better. Notably, our proposed method shows the best performance, with $AP_{0.5}$ reaching 71.3% and 67.8%, respectively, while the "Maximum" method achieved $AP_{0.5}$ of 69.3% and 66.9%. The experimental results indicate that the concentration assessment strategy outperforms the traditional maximum value assessment method in evaluating the effectiveness of response points, aligning with the theoretical analysis of (1).

The comparison between nonhierarchical and hierarchical results indicates that the importance of the five response layers varies, initially revealing that the impacts of old class knowledge revision differ across different depth layers. Additionally, the performance decline of the concentration assessment method in nonhierarchical approaches suggests that the effectiveness of different depth layers cannot be accurately measured using spatial dimension-based evaluation methods. This indirectly indicates the rationality and necessity of considering interlayer relationships (i.e., depth dimension) when evaluating the effectiveness of knowledge in each layer.

It is also worth mentioning that the two submethods in the DFDL approach are both embeddable for any incremental detection method based on hierarchical spatial enhancement. The performance advantage of the hierarchical method over the nonhierarchical method indicates that our DFDL method has a reliable application foundation.

In addition, we analyzed the impact of the hyperparameter p on the performance of the SDDL module. We trained the method using $p = [0.8,\ 0.85,\ 0.9,\ 0.95]$ on each dataset, with incremental tasks set to $2+2$ and $4+3$, respectively. The final training results are shown in Fig. 5. As observed, our method exhibits good stability with respect to the hyperparameter p. Among these, the performance at $p = 0.9$ was found to be the most optimal overall. Therefore, we used 0.9 for the hyperparameter p in subsequent experiments.

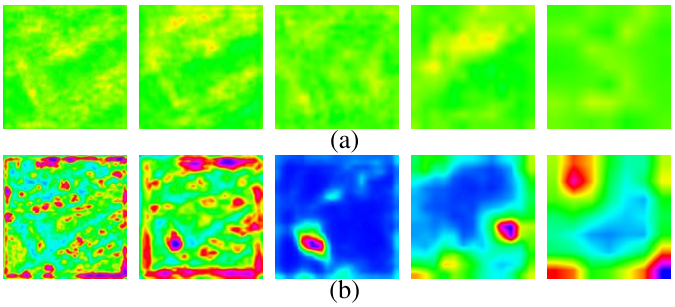


Fig. 6. Heatmap display for two approaches: (a) SA and (b) DFD. Each row represents attention heatmap layers d_0 to d_4 from left to right. Darker red indicates higher heat values at that location, while green and blue indicate low heat values.

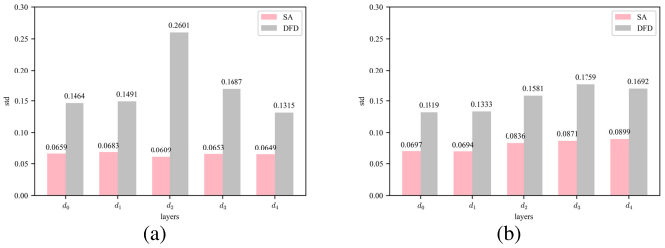


Fig. 7. Standard deviation statistics for heatmap values of different approaches from layers d_0 to d_4. (a) MSAR-1.0. (b) SAR-AIRcraft-1.0.

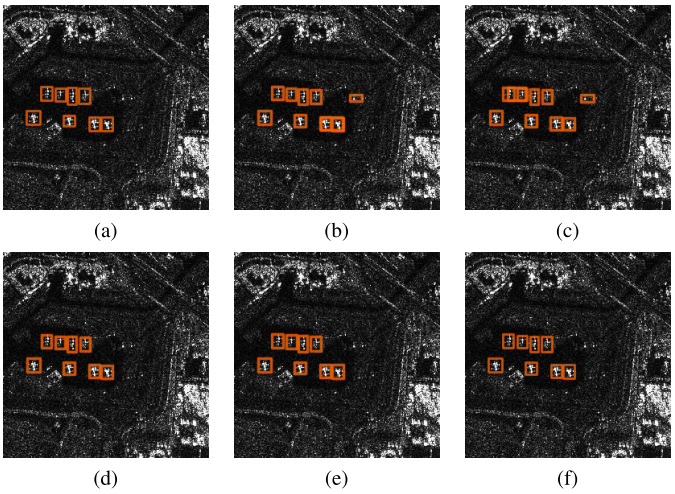


Fig. 8. Comparison of experimental examples of ablation methods: (a) ground truth, (b) SDDL, (c) SDDL+SA, (d) SDDL+DFD, (e) SDDL+HRF, and (f) Ours: SDDL+DFD+HRF. The comparison results show that both the DFD and HRF methods reduce the false alarm rate. At the same time, the "SDDL+DFD+HRF" method performs the best, with the minimal overlap between bounding boxes and more refined localization.

2) Effectiveness of DFDL: In this section, we conducted effectiveness validation of the DFDL submethods based on the SDDL experiment. In the recorded experimental results, "SDDL" represents the smooth distribution distillation method, "DFD" represents the DFD method, "HRF" represents the hierarchical response focal method, "SA" indicates the use of the traditional spatial attention mechanism for feature layer's knowledge distillation instead of the DFD distillation method, and "Ours" represents the complete method SDDFD proposed in this article, which is based on smooth distribution and

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depth-focused distillation. The experimental results are shown in Tables IV and V.

From the experimental results, it is evident that both submethods of the depth-focused approach enhance the model's detection performance compared to the "SDDL" method. Notably, the combined use of both submethods yields the highest performance, achieving $AP_{0.5}$ values of 77.5% and 70.9% on the MSAR-1.0 and SAR-AIRcraft-1.0 datasets, respectively. Moreover, due to the increased focus on localization knowledge by the DFDL method, the complete method also shows improved performance under high localization requirements, with $AP_{0.75}$ values reaching 55.9% and 44.5%, and $AP_{0.5:0.95}$ values reaching 49.2% and 42.8%.

The performance comparison between "SDDL" and "SDDL+HRF" indicates that our HRF method can enhance the performance of hierarchical spatial dimension-based distillation methods, validating the effectiveness of our approach in distinguishing the importance of different depths.

Furthermore, we compared the DFD method with the traditional spatial feature attention method SA. As shown, "SDDL+DFD" achieved significant improvements over the "SDDL" method, whereas the performance of the "SDDL+SA" method declined. This not only demonstrates the effectiveness of our method in decoupling important knowledge from the depth dimension but also indicates that the "SA" method introduced excessive noise, leading to negative impacts. To validate this observation, we examined the attention heatmaps of the FPN's five layers for both methods, as shown in Fig. 6. The color intensity indicates the strength of the attention. We believe that the more distinct the intensity relationship in the attention region, the stronger the method's ability to distinguish effective knowledge in that area. It can be seen that the "SA" method has a weaker differentiation capability in SAR images. To further validate the heatmap observations, we used standard deviation to measure the differentiation degree of each layer's heat values. The calculation method for the standard deviation is as follows:

$$\text{std} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n}} \tag{20}$$

where n is the total number of samples, and $\overline{x}$ is the sample mean. The average standard deviation values for each layer across the two datasets are shown in Fig. 7, which corroborate the analysis from Fig. 6. These results highlight the advantages of our method in the SAR domain.

In addition, we present some experimental examples of various ablation methods in Fig. 8. It can be observed that the "SDDL+SA" method performs the worst, with false alarms and overlapping bounding boxes. Our complete method achieves the best detection performance, with more precise bounding box localization. This demonstrates that our DFDL method is indeed effective in enhancing the model's localization capability.

3) Different Value of λ: We tested the impact of the hyperparameter λ on the performance of our method in the one-step incremental task. The hyperparameter λ is used to balance the volume of distillation loss and ground-truth loss during training, and it is typically less than 1. In this experiment, we tested the performance with λ = [0.10, 0.25, 0.40, 0.55, 0.70]. The $AP_{0.5}$ results for the 2 + 2 task on the MSAR-1.0 dataset and the 4 + 3 task on the SAR-AIRcraft-1.0 dataset are shown in Fig. 9.

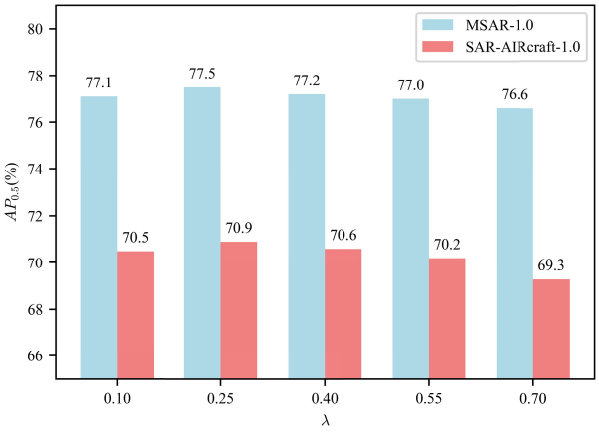


Fig. 9. Experimental results under different values of hyperparameter λ. In these values, the performance of our method is highest when λ = 0.25.

From the results, it can be observed that the performance remains relatively stable and decreases as the value of λ increases. Considering the performance of λ on both datasets, the best results were achieved with λ = 0.25. Therefore, in all other experiments in this article, λ is set to 0.25.

D. Comparison With State-of-the-Art Method

To validate the advantages of our proposed method, we selected four advanced one-stage incremental detection methods for comparison. The methods include RILOD [21], SID [18], ERD [19], and Tian [20], all of which are designed for one-stage detectors. Specifically, the original detector for RILOD is RetinaNet [44], and for SID, it is CenterNet [45]. To ensure consistent evaluation standards and avoid performance upper bound differences caused by varying detection networks, we adapted the RILOD and SID methods to be based on the GFL [38] FCOS [43] network, thereby maintaining consistency in the detection framework across all comparison methods to ensure fairness. Since the earlier ILOD [16] and Faster ILOD [22] methods are designed for two-stage networks, they cannot be transplanted to one-stage networks and were therefore excluded from the comparison.

Our comparative experiments are conducted in both one-step and multistep incremental tasks, and the results are presented in two parts.

In the subsequent experimental results tables, AP_α(old) and AP_α(new) represent the AP_α detection performance of the incremental model on the old and new classes of SAR targets, respectively. Additionally, we recorded the performance gap of each method compared to "Joint Training," with lower values indicated by "−" and higher values indicated by "+." The performance gap is an important and intuitive metric for evaluating the effectiveness of incremental methods. To facilitate observation, we highlight the best results among all methods excluding Joint Training and Fine-Tune, marking the highest AP in red and the smallest "Gap" in blue.

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TABLE VI
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE MSAR-1.0 DATASET WITH A 2 + 2 TASK

Method	Training strategy	$AP_{0.5}$	$AP_{0.5}(old)$	$AP_{0.5}(new)$	$AP_{0.75}$	$AP_{0.75}(old)$	$AP_{0.75}(new)$	$AP_{0.5:0.95}$
Joint Training	Uper Bound	83.7	96.0	71.6	62.3	87.3	37.3	55.8
Fine-Tune	Fine-Tune	34.7	8.3	61.1	19.7	2.3	37.1	19.4
	Gap	−49.0	−87.7	−10.5	−42.6	−85.0	−0.2	−36.4
RILOD [21]	Incremental	63.8	80.4	47.1	34.8	47.7	22.0	33.8
	Gap	−19.9	−15.6	−24.5	−27.5	−39.6	−15.3	−22.0
SID [18]	Incremental	65.6	83.9	47.2	42.2	57.9	26.5	39.6
	Gap	−18.1	−12.1	−24.4	−20.1	−29.4	−10.8	−16.2
ERD [19]	Incremental	68.7	82.5	55.1	44.0	58.9	29.1	41.3
	Gap	−15.0	−13.5	−16.5	−18.3	−28.4	−8.2	−14.5
Tian [20]	Incremental	70.8	86.7	54.9	45.1	59.6	30.7	42.7
	Gap	−12.9	−9.3	−16.7	−17.2	−27.7	−6.6	−13.1
Ours	**Incremental**	**77.5**	**90.2**	**64.8**	**55.9**	**75.7**	**36.1**	**49.2**
	Gap	**−6.2**	**−5.8**	**−6.8**	**−6.4**	**−11.6**	**−1.2**	**−6.6**

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE SAR-AIRCRAFT-1.0 DATASET WITH A 4 + 3 TASK

Method	Training strategy	$AP_{0.5}$	$AP_{0.5}(old)$	$AP_{0.5}(new)$	$AP_{0.75}$	$AP_{0.75}(old)$	$AP_{0.75}(new)$	$AP_{0.5:0.95}$
Joint Training	Uper Bound	73.8	75.3	71.9	46.9	47.2	46.6	45.3
Fine-Tune	Fine-Tune	34.0	7.6	69.2	20.3	1.4	45.5	18.8
	Gap	−39.8	−67.7	−2.7	−26.6	−45.8	−1.1	−26.5
RILOD [21]	Incremental	63.5	67.3	58.4	38.1	42.0	33.0	36.3
	Gap	−10.3	−8.0	−13.5	−8.8	−5.2	−13.6	−9.0
SID [18]	Incremental	64.2	74.0	51.1	37.0	43.7	28.1	36.5
	Gap	−9.6	−1.3	−20.8	−9.9	−3.5	−18.5	−8.8
ERD [19]	Incremental	66.9	73.4	58.2	39.4	44.6	32.6	39.2
	Gap	−6.9	−1.9	−13.7	−7.5	−2.6	−14.0	−6.2
Tian [20]	Incremental	67.3	74.4	57.8	40.6	45.2	34.4	40.1
	Gap	−6.5	−0.9	−14.1	−6.3	−2.0	−12.2	−5.2
Ours	**Incremental**	**70.9**	**74.7**	**65.9**	**44.5**	**47.0**	**41.1**	**42.8**
	Gap	**−2.9**	**−0.6**	**−6.0**	**−2.4**	**−0.2**	**−5.5**	**−2.5**

1) One-Step: For the MSAR-1.0 dataset, we selected ships and oil tanks as the old classes, and airplanes and bridges as the new classes to validate the effectiveness of our method in scenarios where there are significant scale differences between SAR target classes. Table VI shows the experimental results for the 2 + 2 one-step incremental task on the MSAR-1.0 dataset. From the results, it is evident that our method shows a significant performance advantage compared to other advanced methods. Compared to the performance upper bound, the $AP_0.5$ of RILOD, SID, ERD, and Tian decreased by 19.9%, 18.1%, 15.0%, and 12.9%, respectively, while our method only decreased by 6.2%. Specifically, for old and new classes, our method also performed the best, with $AP_0.5$(old) and $AP_0.5$(new) decreasing by 5.8% and 6.8%, respectively. This demonstrates the superiority of our method in scenarios with significant scale differences between SAR target classes.

We also conducted a 4 + 3 incremental task on the SAR-AIRcraft-1.0 dataset, and the results are shown in Table VII. Since all target classes in this dataset are aircraft, the interclass scale differences are smaller, so our method's performance advantage is not as significant as on the MSAR-1.0 dataset. However, it still outperformed all other incremental methods in all metrics. Compared to the performance upper bound, the $AP_0.5$ of RILOD, SID, ERD, and Tian decreased by 10.3%, 9.6%, 6.9%, and 6.5%, respectively, while our method only decreased by 2.9%. Specifically, for old and new classes, our method showed the best performance, with $AP_0.5$(old) and $AP_0.5$(new) decreasing by 0.6% and 6.0%, respectively.

Additionally, the results in Tables VI and VII indicate that our method also significantly improves detection performance under high-precision localization. On the two datasets, our method achieved $AP_0.75$ values of 55.9% and 44.5%, and $AP_{0.5:0.95}$ values of 49.2% and 42.8%, respectively, all of

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TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE MSAR-1.0 DATASET WITH A 2 + 1 + 1 TASK

Method	Training strategy	Step1(%) : 2 + 1			Step2(%) : (2 + 1) + 1		
		$AP_{0.5}$	$AP_{0.5}(old)$	$AP_{0.5}(new)$	$AP_{0.5}$	$AP_{0.5}(old)$	$AP_{0.5}(new)$
Joint Training	Uper Bound	85.0	79.3	96.3	83.7	85.1	79.7
Fine-Tune	Fine-Tune	47.9	25.3	93.1	23.2	5.6	75.8
	Gap	−37.1	−54.0	−3.2	−60.5	−79.3	−3.9
RILOD [21]	Incremental	72.1	65.4	85.4	66.6	66.7	66.2
	Gap	−12.9	−13.9	−10.9	−17.1	−18.4	−13.5
SID [18]	Incremental	73.0	67.2	84.6	68.2	68.4	67.4
	Gap	−12.0	−12.1	−11.7	−15.5	−16.7	−12.3
ERD [19]	Incremental	73.7	66.0	89.1	70.2	69.9	70.9
	Gap	−11.3	−13.3	−7.2	−13.5	−13.7	−8.8
Tian [20]	Incremental	74.7	69.7	84.7	71.3	71.4	70.9
	Gap	−10.3	−9.6	−11.6	−12.4	−13.7	−8.8
Ours	**Incremental**	**82.8**	**80.0**	**89.3**	**75.4**	**76.7**	**71.5**
	Gap	**−2.2**	**+0.7**	**−7.0**	**−8.3**	**−8.4**	**−8.2**

TABLE IX
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE SAR-AIRCRAFT-1.0 DATASET WITH A 3 + 2 + 2 TASK

Method	Training strategy	Step1(%) : 3 + 2			Step2(%) : (3 + 2) + 2		
		$AP_{0.5}$	$AP_{0.5}(old)$	$AP_{0.5}(new)$	$AP_{0.5}$	$AP_{0.5}(old)$	$AP_{0.5}(new)$
Joint Training	Uper Bound	71.8	71.6	72.1	73.8	72.8	76.6
Fine-Tune	Fine-Tune	31.4	8.6	65.6	21.7	1.6	71.8
	Gap	−40.4	−63.0	−6.5	−52.1	−71.2	−5.1
RILOD [21]	Incremental	65.6	71.1	57.4	61.7	66.1	50.7
	Gap	−6.2	−0.5	−14.7	−12.1	−6.7	−25.9
SID [18]	Incremental	65.0	71.2	55.7	60.7	65.7	48.0
	Gap	−6.8	−0.4	−16.4	−13.1	−7.1	−28.6
ERD [19]	Incremental	66.2	69.7	61.1	63.1	66.5	54.7
	Gap	−5.6	−1.9	−11.0	−10.7	−6.3	−21.9
Tian [20]	Incremental	66.6	71.8	58.8	63.7	67.2	55.0
	Gap	−5.2	+0.2	−13.3	−10.1	−5.6	−21.4
Ours	**Incremental**	**69.6**	**72.2**	**65.7**	**68.9**	**69.2**	**68.3**
	Gap	**−2.2**	**+0.6**	**−6.4**	**−4.9**	**−3.6**	**−8.3**

which are the best. This further validates the effectiveness of our method in enhancing the model's localization capability.

Moreover, observing the results across the two datasets reveals a common issue with previous advanced methods: the AP drop ("Gap") relative to the performance upper bound is much more severe for new classes than for old classes. This suggests that these methods, to maintain old class performance, introduce excessive old class constraints during knowledge distillation, thereby hindering new class learning. Our method effectively mitigates this issue, significantly improving new class learning, as evidenced by the smaller difference between "Gap" in AP_α(new) and AP_α(old).

We present some of the experimental results for the one-step incremental tasks on the two datasets, as shown in Figs. 10 and 11. It can be seen that our method's detection capability is the closest to the ground truth. Compared to other incremental detection methods, our method shows fewer false detections and missed detections, and it demonstrates more precise localization for densely packed small targets.

2) Multistep: Tables VIII and IX show the multistep incremental task results on the MSAR-1.0 and SAR-AIRcraft-1.0 datasets, with task settings of 2 + 1 + 1 and 3 + 2 + 2, respectively. In the MSAR-1.0 dataset, due to the insufficient number of images in the airplane class, training solely on this single class would result in small sample issues that are unrelated to the scope of this study, thereby interfering with our observations. Therefore, in this section, we set the initial

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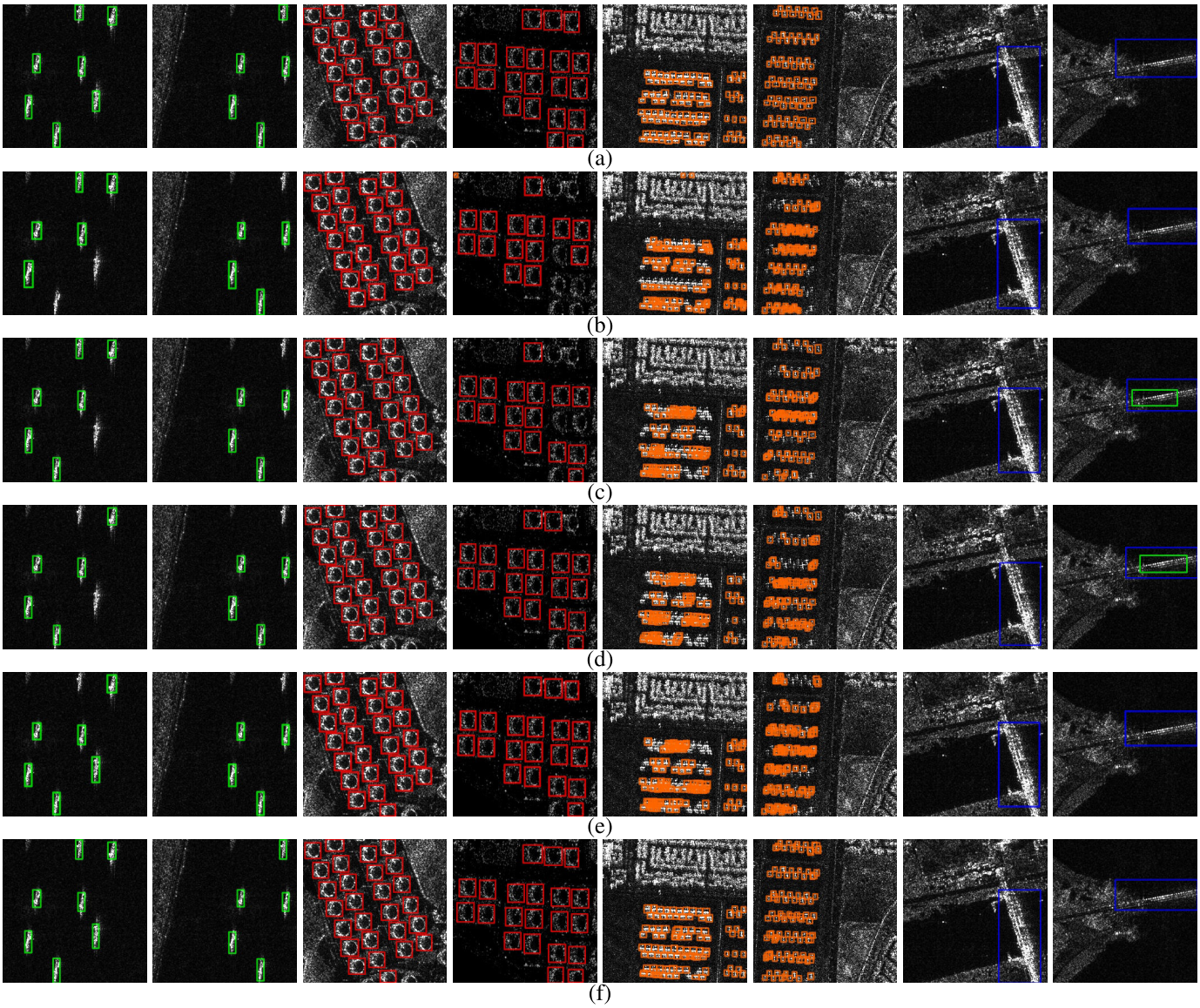


Fig. 10. Comparison of detection results on the MSAR-1.0 dataset. Each row represents a different method: (a) ground truth, (b) RILOD, (c) SID, (d) ERD, (e) Tian, and (f) Ours. Due to the small size and dense arrangement of some targets in this dataset, displaying class labels on the detection bounding boxes would obstruct the view. Therefore, we use green, red, orange, and blue to represent the detection bounding boxes for ships, oil tanks, airplanes, and bridges, respectively, for a clearer observation of the results. From the images, it can be seen that our method has the lowest miss detection rate and performs significantly better than other incremental detection methods in detecting densely arranged small targets.

old classes as ships and airplanes, and incrementally added new classes, starting with oil tanks and then bridges.

From the detection results, it is evident that our method performed the best on both datasets. In the first step incremental task, our method achieved $AP_0.5$ values of 82.8% and 69.6% on the two datasets, respectively, with minimal performance loss for both old and new classes. In the second step incremental task, the $AP_0.5$ values of our method relative to the upper bound, respectively, decreased by 8.3% and 4.9% on the two datasets, still achieving the best performance. This demonstrates the effectiveness of our method in multistep incremental tasks.

In addition, in the multistep incremental experiment results, we observed some external phenomena. We marked the instances with boxes where the incremental methods' performance on old classes exceeded the upper bound of Joint Training. This indicates that the performance of Joint Training is merely a reference for the upper bound and not a strict theoretical limit. Incremental detection methods can outperform Joint Training in certain categories.

Overall, the results of both one-step and multistep incremental experiments demonstrate the effectiveness of our method in class-incremental detection of SAR targets. Our method consistently achieved the best detection performance over other advanced incremental detection methods across various incremental tasks on different datasets.

V. Discussion

In Section IV, we validated the effectiveness of the proposed method. Our method outperformed advanced incremental detection methods across multiple incremental tasks. Nevertheless, there are still several issues worthy of discussion.

First, to avoid the impact of the small sample issue on the experimental results, we balanced the few-shot class of

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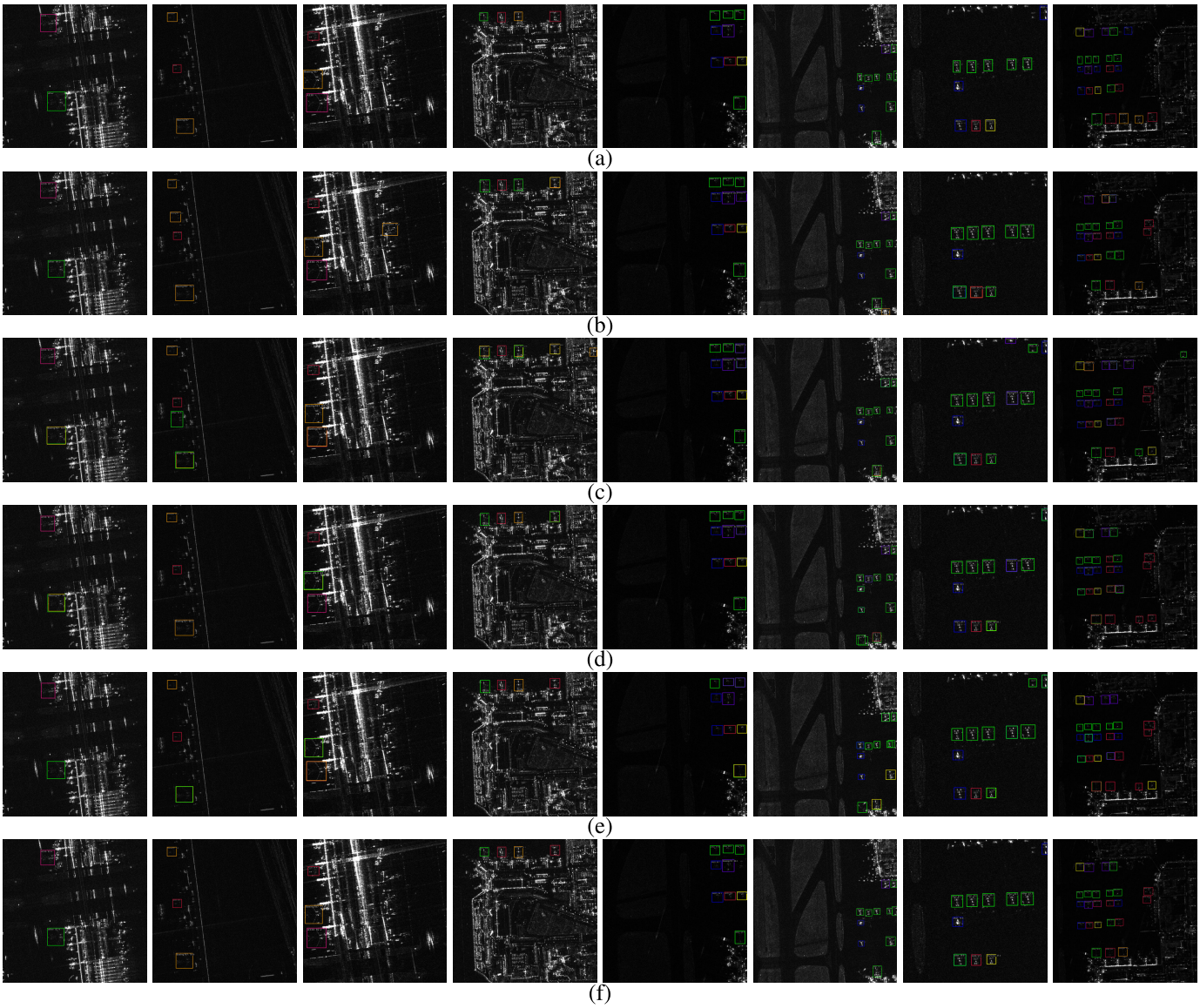


Fig. 11. Comparison of detection results on the SAR-AIRcraft-1.0 dataset. Each row represents a different method: (a) ground truth, (b) RILOD, (c) SID, (d) ERD, (e) Tian, and (f) Ours. In the detection images, different classes of aircraft targets are represented by detection bounding boxes in different colors to facilitate observation and comparison of the results. From the images, it can be seen that our method has lower false-positive and false-negative rates compared to other incremental detection methods.

airplanes in the MSAR-1.0 dataset for this experiment. However, in future application scenarios, small sample incremental detection is an unavoidable research direction. Currently, several studies in the SAR field have addressed the sample imbalance problem [46], [47], but they have not yet been combined with incremental detection methods. How to improve model performance when new class samples are limited is a problem that we need to consider in our future work in few-shot incremental detection.

Second, the one-step experimental results in Section IV-D1 show that previous advanced methods suffer significant performance loss in new class learning. This highlights the conflict between old class constraints and new class knowledge, while our method achieves a better balance. Exploring the boundaries of this conflict and further optimizing the model's incremental detection performance for both old and new classes is a promising direction for future research.

Third, the regression representation method based on GFL [38] provides effective technical support for measuring localization knowledge. In the future, the GFL method may help overcome the challenge of selecting representative old-class localization samples, thus advancing the research of SAR incremental detection methods based on experience replay. Currently, experience replay-based methods have been widely applied in deep reinforcement learning [48] and incremental classification [33], [34], but there has been no research in the field of SAR target incremental detection.

Finally, it should be noted that we have experimentally validated the effectiveness of the idea of differentiating the importance of knowledge from the depth dimension in the DFDL method, and we are sharing this idea to provide a new perspective for researchers in the field of incremental detection. However, due to the interpretability issues of CNNs, our conclusion regarding the impact of depth dimension

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on knowledge is based on our experimental results and is speculative. Currently, our method for evaluating knowledge effectiveness from the depth dimension is still in a relatively qualitative exploratory stage, and further experiments to find more precise and quantitative depth evaluation methods will be a major research direction in the future.

VI. Conclusion

This article proposes a smooth distribution and depth-focused distillation-based class-incremental learning method for SAR target detection to overcome potential challenges posed by the characteristics of SAR imagery. To address the issue of significant background distribution differences in SAR images, we propose a smooth distribution distillation method, which uses a concentration function to evaluate and weigh the effectiveness of distillation regions, thereby suppressing noise interference. To tackle the problem of large-scale differences between different SAR target classes, we propose a depth-focused distillation method, which emphasizes the important locations of localization and classification knowledge from the depth dimension, thereby enhancing the model's localization capabilities for various targets.

We set up multiple incremental learning scenarios on the MSAR-1.0 and SAR-AIRcraft-1.0 datasets. The experimental results show that our method outperforms all other approaches in all scenarios, achieving significant improvements in detection performance. To the best of our knowledge, this is the first incremental detection method that distinguishes knowledge effectiveness from the depth dimension. However, our method's assessment of depth-dimensional effectiveness is still in a relatively qualitative exploratory stage, and developing more precise quantitative assessment methods will be an important direction for future research.

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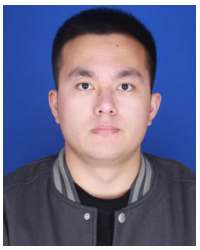
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